CELL LINEAGE FROM SHAPE AND WHOLE-SEQUENCE EVIDENCE

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A PROPOSAL FOR THE BIOHUB CELL TRACKING TASK
WITH THE TEST THAT DECIDES EACH PART

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ABSTRACT

Tracking cells through a developing embryo means detecting every nucleus in every frame of a light-sheet recording and joining the detections into a lineage graph, with each division a fork. This research paper proposes a tracker with no tuned parameter for the Biohub Cell Tracking During Development task. It rests on three hypotheses, each stated with the measurement that would refute it. First, a cell's low-degree spherical harmonic shape energies change less from frame to frame than they differ between neighboring cells, and so carry identity. Second, a division or a death shows as a departure in a record of change taken over the whole recording, which settles after a division and not after a death. Third, an event displaces the cells around it with a far field that points back to it, as a force dipole does in Stokes flow. The hypotheses supply costs to one min-cost flow over the whole recording. None of them is tested here.

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CELL LINEAGE FROM SHAPE AND WHOLE-SEQUENCE EVIDENCE

This research paper proposes a tracker for the Biohub Cell Tracking During Development task, set by its organizers [1], and states for each part of it the measurement that would refute it. Nothing in it is a result. Which parts are built, and what has been measured on them, is recorded in the cell tracking workbook [12].

file	what it holds
tracking_problem.md	the data, the output graph, the score, and the methods the proposal is set against
proposal.md	the constraint the tracker must satisfy, its three hypotheses with the test that decides each, the solve they feed, and what is built

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THE TRACKING PROBLEM

2.1 THE DATA AND THE OUTPUT

Each sample is a light-sheet recording of a developing zebrafish embryo with fluorescently labeled nuclei. Light-sheet microscopy of this type records the nuclei of a whole embryo over hours of development, as Keller et al. [6] showed for the zebrafish. A sample is an OME-Zarr array of shape (T, Z, Y, X): 100 frames of 64 × 256 × 256 voxels, sampled at 1.625 µm in Z and 0.40625 µm in Y and X, as the organizers [1] specify.

The answer is a graph in the Graph Exchange File Format [3]. Each node is one cell in one frame, placed at its center (t, z, y, x). An edge joins a cell to itself in the next frame, and a division is one node with edges to two nodes in the next frame.

2.2 THE SCORE

Under the organizers' [1] metric, predicted nodes are matched to annotated nodes by centroid distance, up to 7 µm. A predicted edge is a true positive when both its ends match annotated nodes that an annotated edge joins. The edge Jaccard index is $J = TP/(TP + FP + FN)$. It is reduced when more nodes are predicted than there are cells:

$$J_{\text{adj}} = \max\left(0,\ J\left(1 - a\,\frac{T_{\text{pred}} - T_{\text{true}}}{T_{\text{true}}}\right)\right), \quad a = 0.1,$$

where T_{pred} is the number of predicted nodes and T_{true} the estimated number of cells. The division Jaccard index counts a predicted fork within one frame of an annotated division as a match. The score is $J_{\text{adj}} + 0.1\,J_{\text{div}}$, micro-averaged over the recordings.

Two consequences follow. A tracker that detects pieces of cells instead of cells loses twice, once in edges that match nothing and once in the node penalty. Divisions carry a tenth of the weight of edges. The Cell Tracking Challenge scores a lineage differently, by the weighted number of graph edits that turn the predicted graph into the annotated one, as defined by Matula et al. [8]. Both score the lineage as a graph, and Ulman et al. [13] compare 21 trackers on the Challenge's measures.

2.3 THE METHODS IT IS SET AGAINST

Current trackers detect cells and then join the detections by one optimization over the whole recording. One form is a min-cost flow in which a division is a node that sends two units of flow, as in Haubold et al. [4]. Another chooses among candidate segmentations drawn from several hierarchies with an integer linear program, as Ultrack [14] does. The proposal retains the whole-recording solve. It changes the evidence the solve is given: each cell's shape and a record of change over the whole recording, in place of costs with weights set by hand or learned.

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LINEAGE FROM SHAPE AND WHOLE-SEQUENCE EVIDENCE

3.1 THE CONSTRAINT

The tracker has no tuned parameter. Every threshold a decision needs is read from the same sample. A detection is retained where it is above what the same procedure finds in frames where no correspondence exists, and a link is costed against the distribution of link lengths in that sample. The constraint excludes a learned model and any weight chosen on withheld data, and it makes each decision traceable to the voxels it came from.

3.2 HYPOTHESIS 1: A SHAPE FINGERPRINT CARRIES IDENTITY

A cell's surface, mapped onto the unit sphere as Brechbühler et al. [2] describe, expands in spherical harmonics, $r(\theta,\varphi) = \sum_{\ell}\sum_{m} c_{\ell m} Y_{\ell m}(\theta,\varphi)$. Medyukhina et al. [9] classify migrating cells with the expansion, by their shape and by how it changes over time. The energy in each degree, $E_{\ell} = \sum_{m} |c_{\ell m}|^2$, does not change when the cell rotates. Kazhdan et al. [5] use this property to match shapes without aligning them, and it compares one cell across frames in the same way.

The lowest degrees have a direct reading. Expanded about its centroid, a shape that is symmetric under inversion through that point, such as any ellipsoid, has no odd-degree terms, because $Y_{\ell m}(-u) = (-1)^{\ell} Y_{\ell m}(u)$. Degree 2 is then the leading anisotropy: it carries the direction of the long axis and how far the shape is flattened or elongated. Degree 1 measures asymmetry about the center, such as a protrusion to one side. Without a segmented surface, the mass, centroid and six second moments of a detected body carry the multipole expansion of its mass to degree 2, together with its radial second moment, and this is the form the workbook builds.

The hypothesis is that a cell's low-degree energies change less from one frame to the next than they differ between neighboring cells, so that they can decide between link candidates that position alone leaves tied. The orthogonality of the $Y_{\ell m}$ does not supply this. It makes the coefficients of one surface unique, and it says nothing about whether the coefficients of two cells differ.

Refutation. Over annotated tracks, take the distance between a cell's fingerprints in consecutive frames, and the distance from its fingerprint to that of its nearest neighbor in the next frame. If the two distributions are no more separated than two draws from one distribution, the fingerprint carries no identity and its cost term is dropped.

3.3 HYPOTHESIS 2: AN EVENT SHOWS IN A WHOLE-SEQUENCE RECORD

A cell commonly divides along its long axis, a rule Minc and Piel [10] review. A division is then expected as two daughters parting along the degree-2 axis of the parent, with their masses summing to the parent's. The hypothesis adds evidence from the whole recording. A record of change over the whole sample, built in the workbook as the count of flips of each bit of each voxel's intensity, departs from its own background within one frame of each division. After a division the record settles; after a death it does not. A track that ends with no such departure is then a missed detection, not a death, and a fork with none is a false division.

Refutation. If the record's departure at annotated division frames is no larger than at the same number of frames drawn at random from the same samples, the record carries no event signal.

3.4 HYPOTHESIS 3: AN EVENT DISPLACES ITS NEIGHBORS, AND THE DISPLACEMENT POINTS BACK TO IT

A dividing or dying cell moves the tissue around it. At the length and speed of cells the Reynolds number is far below one, as Purcell [11] sets out. If the tissue responds as a viscous fluid, the flow around an event is a Stokes flow. A body that exerts no net force on its surroundings has a far field led by a force dipole, whose velocity falls off as $1/r^2$, followed by a source dipole falling off as $1/r^3$, as Lauga and Powers [7] review for swimming microorganisms. About the event, the force dipole's flow has the angular structure of degree 2 and the source dipole's that of degree 1. The neighbors' departures from the common drift are therefore expanded in those degrees about each candidate origin, and the origin that best accounts for them is taken as the event's position.

Embryonic tissue is packed with cells and is viscoelastic instead of a Newtonian fluid. The Stokes far field is a hypothesis about it instead of a property of it.

Refutation. Take the mean magnitude of the neighbors' departures from the common drift at distance r from annotated divisions, and the same mean at distance r from points away from any division. If the first is no larger than the second, there is no displacement to read. If it is larger, its fall-off with r tests the Stokes reading, which predicts $1/r^2$ in the far field.

3.5 THE SOLVE

The three hypotheses supply costs to one solve over the whole recording: a min-cost flow over every frame, in which a division sends two units of flow, as in Haubold et al. [4]. Every cost is an exact integer on one footing, and the draws from frames with no correspondence take the place of weights.

3.6 WHAT IS BUILT

The cell tracking workbook [12] records which parts are built and what has been measured on them. The second-moment fingerprint and the whole-sample record of bit flips are built. A one-to-one linker between consecutive frames, costed on position and drift alone, is built and scored. The

fingerprint is not yet among the linker's costs. The event tests, the displacement reading and the solve over the whole recording are not built, and none of the three hypotheses has been tested.

3.7 DATA AND CODE AVAILABILITY

The recordings and annotations are distributed through Kaggle under the competition's terms, and the organizers' [1] baseline and metric are public. The tracker's code is in the orior repository, and its measurements are recorded in the cell tracking workbook [12].

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